

①

~~$[x_1^{(i)}, x_2^{(i)}, \dots, x_n^{(i)}, y^{(i)}]_{i \in \mathbb{I}_m}$ Ver sefi olsun.~~

$$f(x_1, x_2, \dots, x_n) = a_1 \cdot x_1 + a_2 \cdot x_2 + \dots + a_n \cdot x_n + b \quad (\text{Linear Func})$$

$[x_1^{(i)}, x_2^{(i)}, \dots, x_n^{(i)}, y^{(i)}]_{i \in \mathbb{I}_m}$ Ver sefi olsun.

$$J(a_1, a_2, \dots, a_n, b) = \frac{1}{m} \sum_{i=1}^m (f(x_1^{(i)}, x_2^{(i)}, \dots, x_n^{(i)}) - y^{(i)})^2$$

$$J(a_1, a_2, \dots, a_n, b) = \frac{1}{m} \cdot \sum_{i=1}^m (a_1 \cdot x_1^{(i)} + a_2 \cdot x_2^{(i)} + \dots + a_n \cdot x_n^{(i)} + b - y^{(i)})^2$$

$\vec{a} = (a_1, a_2, \dots, a_n)$ katsayılar vektörün,

$\vec{x} = (x_1, x_2, \dots, x_n)$ n değıskeri değıskeri vektörün değıskeri.

$j \in \mathbb{I}_n = \{1, 2, \dots, n\}$ olsun örne;

$$\left[\frac{J(a_1, a_2, \dots, a_j + \Delta_j, \dots, a_n, b) - J(a_1, a_2, \dots, a_n, b)}{\Delta_j} \right]_{(\Delta_j \neq 0)} =$$

$$= \frac{1}{\Delta_j} \cdot (J(a_1, a_2, \dots, a_j + \Delta_j, \dots, a_n, b) - J(\vec{a}, b))$$

~~$$= \frac{1}{\Delta_j} \cdot \left(\frac{1}{m} \sum_{i=1}^m (f(x_1^{(i)}, x_2^{(i)}, \dots, x_n^{(i)}) - y^{(i)})^2 - \frac{1}{m} \sum_{i=1}^m (a_1 \cdot x_1^{(i)} + a_2 \cdot x_2^{(i)} + \dots + a_n \cdot x_n^{(i)} + b - y^{(i)})^2 \right)$$~~

$$= \frac{1}{\Delta_j} \cdot \left[\frac{1}{m} \cdot \sum_{i=1}^m (a_1 \cdot x_1^{(i)} + a_2 \cdot x_2^{(i)} + \dots + (a_j + \Delta_j) \cdot x_j^{(i)} + \dots + a_n \cdot x_n^{(i)} + b - y^{(i)})^2 - \frac{1}{m} \cdot \sum_{i=1}^m (a_1 \cdot x_1^{(i)} + a_2 \cdot x_2^{(i)} + \dots + a_n \cdot x_n^{(i)} + b - y^{(i)})^2 \right]$$

$$= \frac{1}{n \cdot \Delta_j} \sum_{j=1}^n \left[\left(a_1 \cdot x_1^{(j)} + a_2 \cdot x_2^{(j)} + \dots + a_j \cdot x_j^{(j)} + \Delta_j \cdot x_j^{(j)} + \dots + a_n \cdot x_n^{(j)} + b - y^{(j)} \right)^2 \right] \quad (2)$$

$$= \frac{1}{n \cdot \Delta_j} \sum_{j=1}^n \frac{\Delta_j \cdot x_j^{(j)} \cdot \left(2 \cdot (\vec{a} \cdot \vec{x}^{(j)} + \Delta_j \cdot x_j^{(j)}) \right)}{\Delta_j \cdot x_j^{(j)} \cdot \left(2 \cdot (\vec{a} \cdot \vec{x}^{(j)} + \Delta_j \cdot x_j^{(j)}) \right)}$$

$$= \frac{1}{n} \cdot \sum_{j=1}^n x_j^{(j)} \cdot \left(2 \cdot (a_1 x_1^{(j)} + a_2 x_2^{(j)} + \dots + a_n x_n^{(j)} + b - y^{(j)}) + \Delta_j \cdot x_j^{(j)} \right)$$

$$= \frac{1}{n} \cdot \sum_{j=1}^n x_j^{(j)} \cdot \left(2 \cdot x_1^{(j)} \cdot a_1 + 2 \cdot x_2^{(j)} \cdot a_2 + \dots + 2 \cdot x_n^{(j)} \cdot a_n + 2 \cdot b - 2 \cdot y^{(j)} + \Delta_j \cdot x_j^{(j)} \right)$$

$$= \frac{1}{n} \cdot \sum_{j=1}^n \left(2 \cdot x_j^{(j)} \cdot x_1^{(j)} \cdot a_1 + 2 \cdot x_j^{(j)} \cdot x_2^{(j)} \cdot a_2 + \dots + 2 \cdot x_j^{(j)} \cdot x_n^{(j)} \cdot a_n + 2 \cdot x_j^{(j)} \cdot b - 2 \cdot x_j^{(j)} \cdot y^{(j)} + (x_j^{(j)})^2 \cdot \Delta_j \right)$$

$$= \left(\frac{2}{n} \cdot \sum_{j=1}^n x_j^{(j)} \cdot x_1^{(j)} \right) \cdot a_1 + \left(\frac{2}{n} \cdot \sum_{j=1}^n x_j^{(j)} \cdot x_2^{(j)} \right) \cdot a_2 + \dots + \left(\frac{2}{n} \cdot \sum_{j=1}^n x_j^{(j)} \cdot x_n^{(j)} \right) \cdot a_n + \left(\frac{2}{n} \cdot \sum_{j=1}^n x_j^{(j)} \right) \cdot b - \left(\frac{2}{n} \cdot \sum_{j=1}^n x_j^{(j)} \cdot y^{(j)} \right) + \left(\frac{1}{n} \cdot \sum_{j=1}^n (x_j^{(j)})^2 \right) \cdot \Delta_j$$

$\Delta_j \rightarrow 0$ $\Delta_j \rightarrow 0$

$$E_1 \cdot a_1 + E_2 \cdot a_2 + \dots + E_n \cdot a_n + E_{n+1} \cdot b - E_{n+2}$$

(3)

$$\boxed{\frac{\partial J(a_1, a_2, \dots, a_n, b)}{\partial a_j} = \left[\varepsilon_1 a_1 + \varepsilon_2 a_2 + \dots + \varepsilon_n a_n + \varepsilon_{n+1} b - \varepsilon_{n+2} \right]}$$

$$= \left(\frac{2}{m} \cdot \sum_{j=1}^m x_j^{(1)} \cdot x_1^{(1)} \right) \cdot a_1 + \dots + \left(\frac{2}{m} \cdot \sum_{j=1}^m x_j^{(n)} \cdot x_n^{(n)} \right) \cdot a_n$$

$$+ \left(\frac{2}{m} \cdot \sum_{j=1}^m x_j^{(n+1)} \right) \cdot b - \frac{2}{m} \cdot \sum_{j=1}^m x_j^{(n+2)} \cdot y^{(j)}$$

$$= \frac{2}{m} \cdot \left[\sum_{j=1}^m (x_j^{(1)} \cdot x_1^{(1)} \cdot a_1) + \sum_{j=1}^m (x_j^{(2)} \cdot x_2^{(2)} \cdot a_2) + \dots + \sum_{j=1}^m (x_j^{(n)} \cdot x_n^{(n)} \cdot a_n) \right. \\ \left. + \sum_{j=1}^m (x_j^{(n+1)} \cdot b) - \sum_{j=1}^m x_j^{(n+2)} \cdot y^{(j)} \right]$$

$$= \frac{2}{m} \cdot \sum_{j=1}^m \left(\underline{x_j^{(1)}} \cdot \underline{x_1^{(1)}} \cdot a_1 + \underline{x_j^{(2)}} \cdot \underline{x_2^{(2)}} \cdot a_2 + \dots + \underline{x_j^{(n)}} \cdot \underline{x_n^{(n)}} \cdot a_n + \underline{x_j^{(n+1)}} \cdot b - \underline{x_j^{(n+2)}} \cdot y^{(j)} \right)$$

$$= \boxed{\frac{2}{m} \cdot \sum_{j=1}^m x_j^{(1)} \cdot \left(a_1 x_1^{(1)} + a_2 x_2^{(1)} + \dots + a_n x_n^{(1)} + b - y^{(1)} \right)}$$

$$= \boxed{\frac{2}{m} \cdot \sum_{j=1}^m x_j^{(1)} \cdot \left(f(x_1^{(1)}, x_2^{(1)}, \dots, x_n^{(1)}) - y^{(1)} \right)}$$

$$= \boxed{\frac{2}{m} \cdot \sum_{j=1}^m x_j^{(1)} \cdot \left(f(\vec{x}^{(1)}) - y^{(1)} \right)}$$

$$\boxed{\frac{\partial J(a_1, a_2, \dots, a_n, b)}{\partial a_j}}$$